numpy is the structer of the python while using the numpy the varible for data.

```
pip install numpy
```

```
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
```

```
import numpy as np
```

```
print(np.__version__)
```

```
2.0.2
```

```
a=np.ndarray([1,2,3,4])
print(a)
```

```
[[[[-0.25 -0.5   0.   -0.25]
   [ 0.25 -0.5   0.5  -0.25]
   [ 0.25  0.    0.5   0.25]]

  [[ 0.25  0.5   0.    0.25]
   [-0.25  0.5  -0.5   0.25]
   [-0.25  0.   -0.5  -0.25]]]]
```

```
a=np.array([1,2,3,4])
print(a)
```

```
[1 2 3 4]
```

```
b=np.array([[1,2,3],
            [4,5,6],
            [7,8,9]
            ])
print(b)
```

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
```

```
#3d array
c=np.array([[[1,2,3],
            [4,5,6],
            [7,8,9]
            ],
            [[1,2,3],
            [4,5,6],
            [7,8,9]
            ]
            ])
print(c)
```

```
[[[1 2 3]
  [4 5 6]
  [7 8 9]]

 [[1 2 3]
  [4 5 6]
  [7 8 9]]]
```

```
a=np.ones([3,3])
print(a)
```

```
[[1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]]
```

```
e=np.zeros([2,3])
print(e)
```

```
[[0. 0. 0.]
 [0. 0. 0.]]
```

```
c=np.full([2,3],9)
print(c)
```

```
[[9 9 9]
 [9 9 9]]
```

```
e=np.eye(4)
print(e)
```

```
[[1. 0. 0. 0.]
 [0. 1. 0. 0.]
 [0. 0. 1. 0.]
 [0. 0. 0. 1.]]
```

```
f=np.arange(2,10,3)
print(f)
```

```
[2 5 8]
```

```
np.linspace(2,10,12)
```

```
array([ 2.        ,  2.72727273,  3.45454545,  4.18181818,  4.90909091,
        5.63636364,  6.36363636,  7.09090909,  7.81818182,  8.54545455,
        9.27272727, 10.        ])
```

```
#Random arrays
np.random.rand(3,3)
```

```
array([[0.51711423, 0.55015791, 0.57217439],
       [0.69276608, 0.79457491, 0.21029699],
       [0.77591121, 0.82233601, 0.86089583]])
```

```
np.random.randint(2,5,size=(4,4))
```

```
array([[2, 4, 2, 2],
       [4, 2, 3, 3],
       [4, 4, 4, 2],
       [4, 3, 2, 3]])
```

```
#unintialized
np.empty((3,5))
```

```
array([[5.41661642e-315, 0.00000000e+000, 3.02876088e-152,
        4.83784398e+276, 7.57857007e-010],
       [9.00495283e+130, 1.28822509e-057, 5.53288060e-048,
        7.67313859e-010, 4.19104394e+175],
       [6.69505074e-109, 1.35717430e+131, 1.05146952e-153,
        1.05190653e-153, 1.42290906e-321]])
```

```
arr=np.array([["a","b","c"],[1,2,3]])
print(arr.shape)
print(arr.ndim)
print(arr.size)
print(arr.dtype)
print(arr.itemsize)
print(arr.nbytes)
```

```
(2, 3)
2
6
<U21
84
504
```

```
a=np.array([10,20,30,40])
print(a[0])
print(a[-1])
```

```
10
40
```

```
arr=np.array([1,2,3,4,55,77])
print(a[1:4])
print(a[2:])
print(a[::3])
print(a[::-1])
```

```
[20 30 40]
[30 40]
[10 40]
[40 30 20 10]
```

```
m=np.array([[1,2],
            [3,4]])
#single elements:[row,col]
```

```
print(m[0,1])
print(m[1,0])
print(m[1,1])
#row slicing
print(m[0,:])
print(m[1,:])
#col slicing
print(m[:,0])
print(m[:,1])
```

```
2
3
4
[1 2]
[3 4]
[1 3]
[2 4]
```

```
#boolean
a=np.array([11,44,56,3])
mask=a>44
print(mask)
print(a[mask])
```

```
[False False  True False]
[56]
```

```
#shorthand
print(a[a>44])
```

```
[56]
```

```
#fancy indexing
print(a[[0,2]])
```

```
[11 56]
```

```
import numpy as np
a=np.array([1,2,3,4])
b=np.array([4,5,6,7])
print(a+b)
print(a-b)
print(a*b)
print(a/b)
print(a**b)
print(a**2)
print(a+10)
```

```
[ 5  7  9 11]
[-3 -3 -3 -3]
[ 4 10 18 28]
[0.25       0.4        0.5        0.57142857]
[    1    32   729 16384]
[ 1  4  9 16]
[11 12 13 14]
```

```
a=np.array([1.0,22.0,33.0,44.0])
np.floor(a)
np.ceil(a)
np.round(a,2)
```

```
array([ 1., 22., 33., 44.])
```

```
Start coding or generate with AI.
```